CAPSTONE PROJECT

SUBMITTED BY: LESLIE ANNE D. DOLOT

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OUTLINE:

- Business Problem
- Our Approach
- Key Findings
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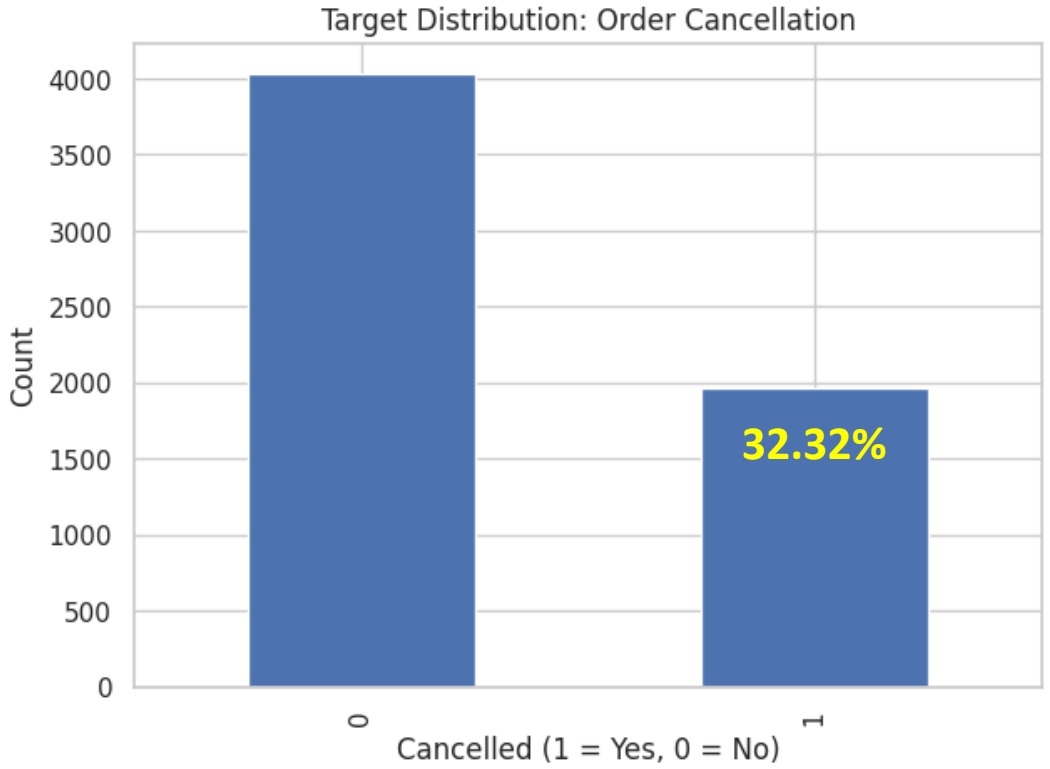
- Recommendations
- Risks and Considerations
- Next Steps & ROI Projection

foodpanda

Business Problem

- There are **32.32%** cancelled orders in the past year from August 2024 to 2025.
- The order cost of cancellations ranges from 100 to 1500 Indian Rupees.
- The months of January, March, August, September and October have the highest cancellations (>500 counts)

We lost 1,571,276.37 Indian Rupees ($ 16,624) due to order cancellations in one year.



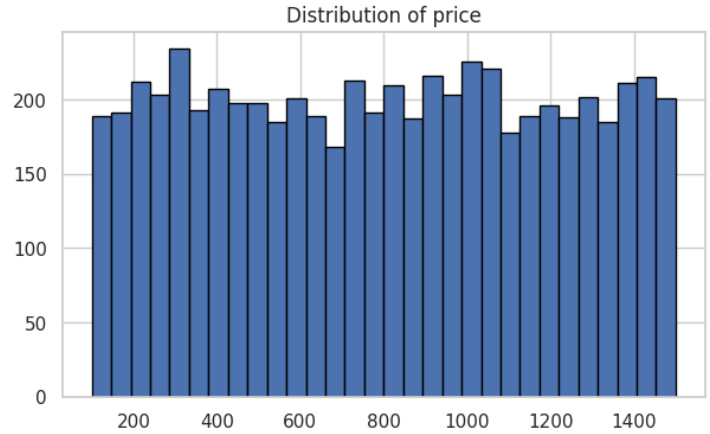
Our Approach

- Identify:

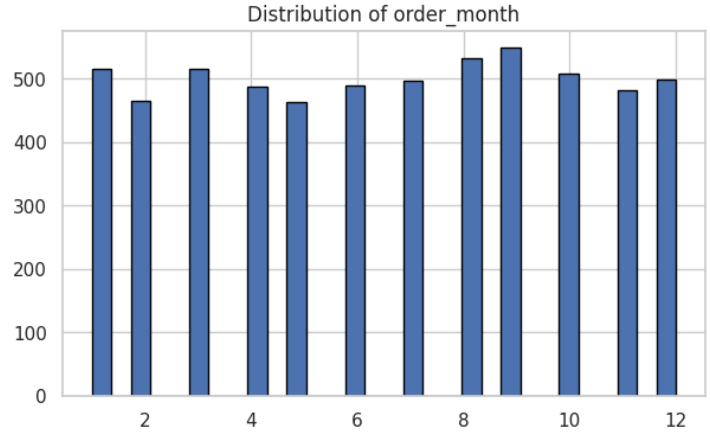
  Which orders are likely to be cancelled before fulfillment?

- Once identified:
  - ✓ Provide proactive interventions (discounts, faster dispatch, alerts)
  - ✓ Create system changes, solutions and preventive measures
  - ✓ Measure the savings and earnings from these interventions
  - ✓ Review customers' brand loyalty and patronage

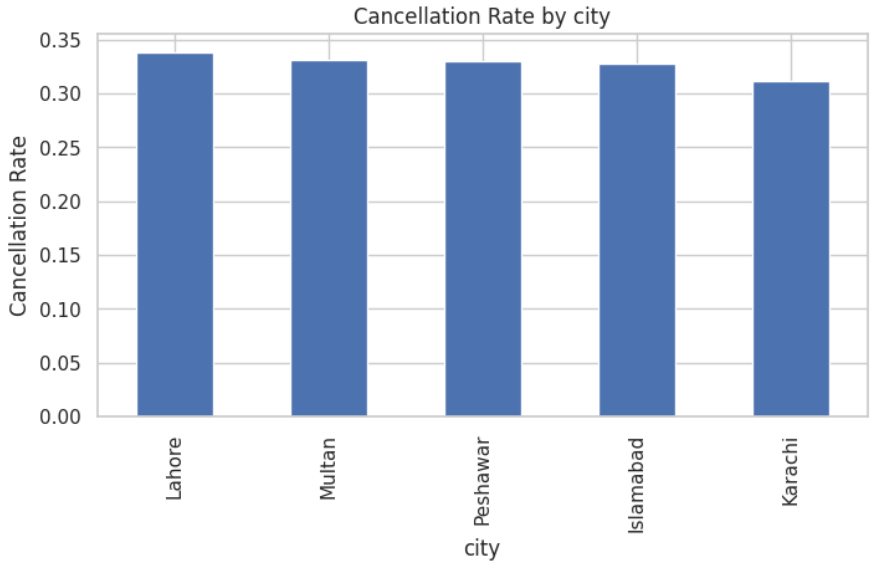
Key Findings



- Cancellation happens across the full range of the transactions' price or amount.



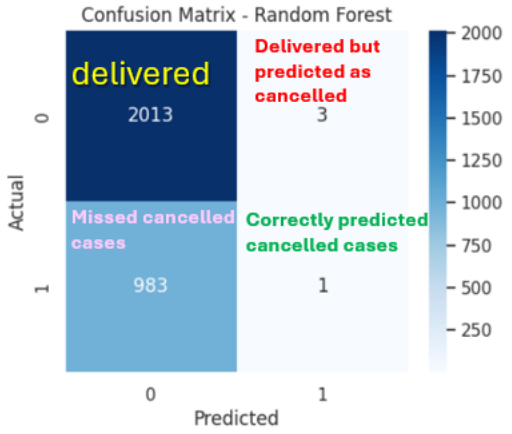
- High cancellations (>500 counts) occurred during the months of January, March, August, September and October.

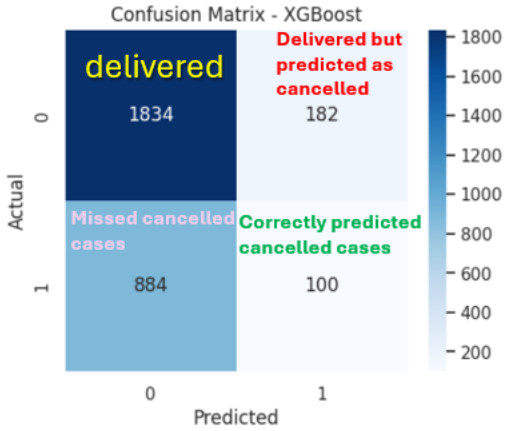


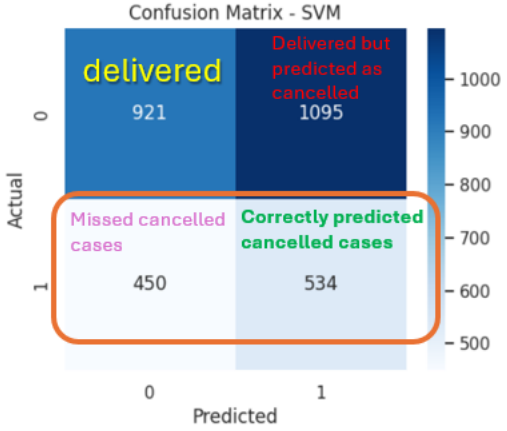
- Certain locations have higher cancellations than others.

Results

- Supervised Classification Algorithm is employed comparing three algorithms: Random Forest, XGBoost, and SVM to determine which model will provide the answer on which features can predict order cancellations.







MODEL COMPARISON, ROC-AUC, PR-AUC, RECALL, PRECISION, F1

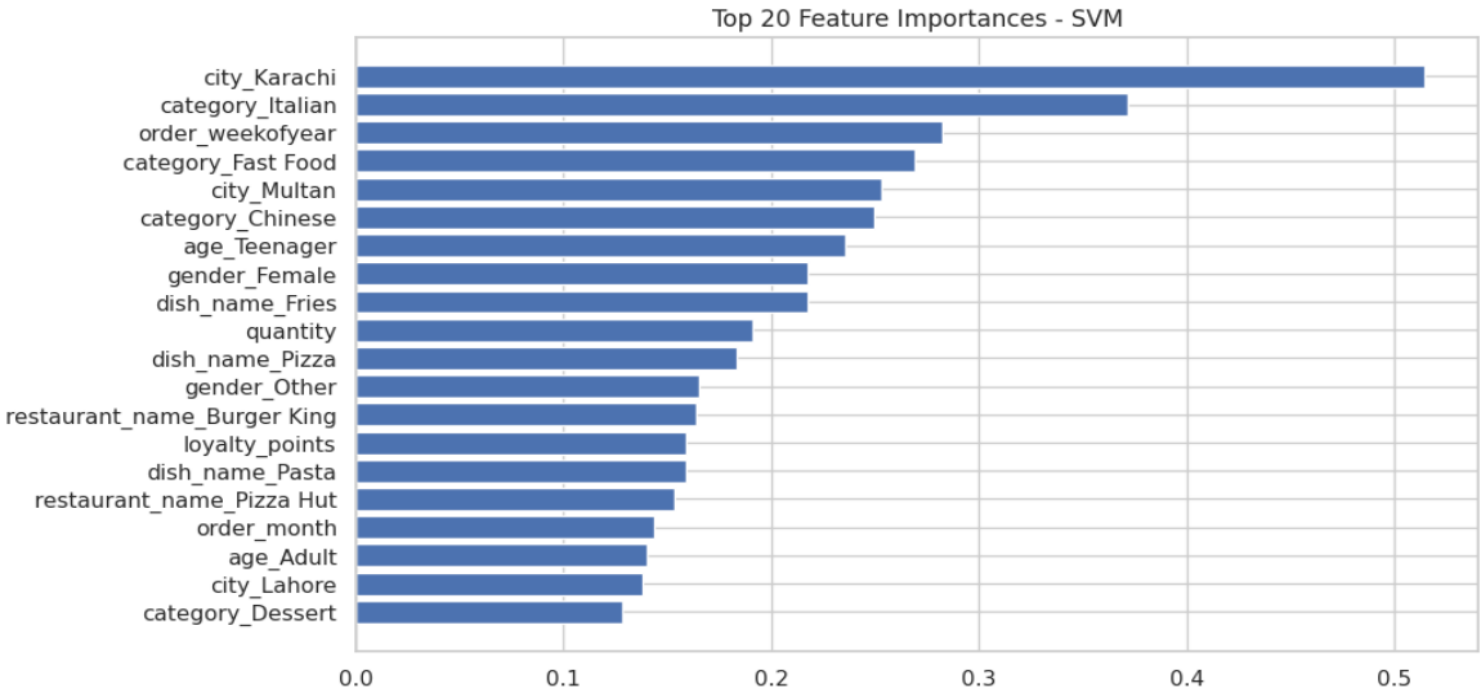
	model	roc_auc	pr_auc	recall	precision	f1_score	accuracy	correct_predictions	total_predictions
0	Random Forest	0.520421	0.340028	0.001016	0.250000	0.002024	0.671333	2014	3000
1	XGBoost	0.508862	0.342794	0.101626	0.354610	0.157978	0.644667	1934	3000
2	SVM	0.501281	0.322063	0.542683	0.327808	0.408726	0.485000	1455	3000

→ Selected Model

Results

Top 20 feature importances - SVM

	feature	importance
0	city_Karachi	0.514801
1	category_Italian	0.371782
2	order_weekofyear	0.282661
3	category_Fast Food	0.269170
4	city_Multan	0.252810
5	category_Chinese	0.249435
6	age_Teenager	0.235590
7	gender_Female	0.217729
8	dish_name_Fries	0.217591
9	quantity	0.191199
10	dish_name_Pizza	0.183187
11	gender_Other	0.165252
12	restaurant_name_Burger King	0.163807
13	loyalty_points	0.159295
14	dish_name_Pasta	0.159254
15	restaurant_name_Pizza Hut	0.153162
16	order_month	0.143957
17	age_Adult	0.139943
18	city_Lahore	0.138092
19	category_Dessert	0.128073



- the order_weekofyear, quantity, and city_Karachi are the top 3 most influential features in the data that can modestly predict cancellation of orders.

Recommendations

- Management may schedule a more assertive interventions (promos or vouchers) during the months where high order cancellations occurs (> 500 counts) to reduce the number of cancelled orders. These are during the months of January, March, August, September, and October.
- Management may provide promos of small quantity orders to encourage customers to order more quantities and reduce the possibility of cancelling the order.
- To review and validate further the observed high cancellation rate on certain cities and confirm whether or not there is bias on the data or if it is real the actual reality.
- To strengthen the predictive power of the model, there should be more data collected answering the following areas:
  - a. app's indicated delivery time
  - b. time between the placement of order and the delivery (or cancellation)
  - c. driver assignment time (how long for the order to be assigned to a driver)
  - d. duration of time for the restaurant to fulfill the order
  - e. reason for order cancellation (availability, waiting time, rush hour, etc.)
  - f. types of cancellations (customer cancels due to change of mind, restaurant cancels due to out of stock, system cancels, or rider cancels)
  - g. other external factors (weather, road works, holidays, etc.)

Risks and Considerations

- Address data imbalance and limitations
  - Some features and subsets have small data population and this may lead to skewing of data, affecting integrity of the predictive power of the model
- Enlarge the dataset and cover other pertinent features
  - Include other features such as delivery time, reasons for cancellation, types of cancellation, and other external factors)
- Watch out for biases and perform fairness audits
  - Based on the demographic parity test conducted, it indicated a significant disparity in predicted cancellation rates across different cities. This disparity can be addressed by adding more data points and other mitigating measures to process data.
- Use of fairness metrics to mitigate biases
- Review mitigations for implementation (Reweighting and post-processing)
- Ensure data privacy while collecting data, and proper management of personal data in repository

Next Steps and ROI Projection

Once interventions are already in place, calculate the following metrics to measure business KPI:

Cancellation Rate Reduction

Formula:

(New Cancellation Rate - Old Rate) / Old Rate

Objective:

Lower overall cancellations

Prevented Cancellations

Formula:

Prevented Cancellations = total orders saved due to intervention

Objective:

Reduce the % of cancelled orders in the same time frame as the previous data.

Revenue Retained

Formula:

Revenue Retained = ∑(value of orders where cancellation is prevented)

Objective:

The higher the value, then more revenue is generated.

Intervention Cost vs Savings

Formula:

Net Benefit = Revenue Retained - Cost of Intervention

Objective:

The Net Benefit must be a positive value.

Thank you!